Aishwarya Balwani

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Education & Training

St. Jude Children's Research Hospital — Memphis, TN
Postdoctoral Research Associate, Department of Developmental Neurobiology — *2025 – Present*

Georgia Institute of Technology — Atlanta, GA
PhD, Electrical & Computer Engineering — *2018 – 2025*

- Minors: Mathematics, Computer Science
- Thesis: *Through the Recurrent Neural Network Looking Glass: Structure-Function Relationships in Cortical Circuits for Predictive Coding*

MS, Electrical & Computer Engineering — *2016 – 2018*

University of Mumbai — Mumbai, India
BE, Electronics & Telecommunication (First Class with Distinction) — *2012 – 2016*

Research & Work Experience

Postdoctoral Research Associate — Fall 2025 – Present
St. Jude Children's Research Hospital — *Advisor: Dr. Stanislav Zakharenko*

- Elucidating neuronal circuit mechanisms underlying auditory hallucinations in mouse models of schizophrenia.
- Developing and using methods from high-dimensional geometry, statistical machine learning, deep learning, and mechanistic interpretability on 2-photon calcium imaging data from the auditory cortex.
- Also working to find behavioural correlates of auditory hallucinations by interpreting pose-tracking models of freely behaving mice.

Interpretability Researcher — Fall 2026
Verified Mechanisms

- Identifying and understanding theoretical results emerging from autoresearch runs to advance the broader goal of characterizing what attention heads compute and how their computations can be understood mechanistically.
- Developing an empirical framework to extract and analyze mathematically-derived atoms learnt by attention heads, investigating their functional signatures, necessity/sufficiency, and redundancies.
- Investigating feature maps and kernel structures induced by these atoms, to understand the geometry of their function spaces and connecting their properties to attention-head complexity and minimal representations.

Researcher & Project Mentor — Fall 2026
Supervised Program on Alignment Research (SPAR)

- Leading multiple projects studying unintended generalization in LLMs through the lens of the Lottery Ticket Hypothesis, Mode Connectivity, and low-rank geometry in the parametric space.
- Proposed the project outline, secured compute funding for the duration of the program, and currently advising 8 mentees ranging from senior high-schoolers to Masters students.

ML Researcher & Research Mentor — Summer 2025 – Present
Algoverse AI Research

- Mentoring high-achieving college students in applied ML and LLM research targeting workshops at top ML/AI conferences.

Technical Advisor — February 2026 – September 2026
SmaLLM

- Advising on ML algorithms, fine-tuning strategies, model architectures, and statistical techniques.

AI Strategy Consultant, Frontier Tech — September 2025 – June 2026
Scale AI

- Developed deep research benchmarks and applied Chain-of-Thought interpretability to evaluate and improve tool use capabilities of agentic LLMs.

Graduate Research Assistant — Summer 2018 – Spring 2025
Georgia Institute of Technology

- Studied predictive coding, incorporating biological constraints in RNNs, representational geometry, and architectural biases in the cortex. (Advisor: Dr. Hannah Choi)
- Studied mutli-scale models of brain structure and organization, representation learning. (Advisor: Dr. Eva Dyer)

Forecasting Mentee Summer 2023

Epoch FRI AI Mentorship Program *Mentor: Molly Hickman*

- Developed Bayesian frameworks to quantify how antecedent questions predict beliefs on complex topics.

Winter Project Intern December 2022 – January 2023

Good Futures Initiative, EA Berkeley

- Developed a mathematical framework using representational geometry to detect and address AI misalignment.

Summer Research Associate Summer 2022

Center for Computational Neuroscience, Flatiron Institute, Simons Foundation *Supervisor: Dr. SueYeon Chung*

- Developed a three-factor Hebbian learning rule for non-negative recurrent networks and analyzed auditory cortex representations in mice.

R&D Intern, Algorithms Team Summer 2017

Intellifusion, China

- Worked on algorithms for image processing, data compression, and encryption.

Publications, Preprints & Peer Reviewed Abstracts

Publications

Balwani A., Wang A. Y., Najafi F., Choi H. "Constructing Biologically Constrained RNNs via Dale's Backpropagation and Topologically-Informed Pruning." *Science Advances*, 2025.

Sharma M., Zhang C., Bandi C., Wang C., ... **Balwani A.**, ... & Liu B. "ResearchRubrics: A Benchmark of Prompts and Rubrics For Evaluating Deep Research Agents." *arXiv*, 2025 (Accepted and to be presented at *ICLR,* 2026).

Balwani A., Cho S., & Choi H. "On the Architectural Biases of the Canonical Cortical Microcircuit." *Neural Computation*, 2025.

Ozan Bozdag G., Zamani-Dahaj S.A., Kahn P., Day T., Tong K., **Balwani A.**, Dyer E., Yunker P., & Ratcliff W. "*De Novo* Evolution of Macroscopic Multicellularity." *Nature*, 2023.

Balwani A., & Krzyston J. "Zeroth-order Topological Insights into Magnitude-based Neural Network Pruning." *PMLR Volume on Topology, Algebra, and Geometry in Learning*, 2022.

Balwani A.*, Miano J.*, Liu R., Kitchell L., Prasad J., Johnson E., Gray-Roncal W., & Dyer E. "Multi-Scale Modeling of Neural Structure in X-ray Imagery." *IEEE International Conference on Image Processing (ICIP)*, 2021.

Prasad J., **Balwani A.**, Johnson E., Miano J., Sampathkumar V., De Andrade V., . . . & Dyer E. "A three-dimensional thalamocortical dataset for characterizing brain heterogeneity." *Nature Scientific Data*, 2020.

Liu R., Subakan C., **Balwani A.**, Whitesell J., Harris J., Koyejo S., & Dyer E. "A generative modeling approach for interpreting population-level variability in brain structure." *MICCAI*, 2020.

Balwani A., & Dyer E. "Modeling variability in brain architecture with deep feature learning." *2019 53rd Asilomar Conference on Signals, Systems, and Computers. IEEE*, 2019.

Milligan K., **Balwani A.**, & Dyer E. "Brain Mapping at High Resolutions: Challenges and Opportunities." *Current Opinion in Biomedical Engineering*, 2019.

Lee T., Kumar A., **Balwani A.**, Brittain D., Kinn S., Tovey C., Dyer E., da Costa N., Reid R., Forest C., & Bumbarger D. "Large-scale neuroanatomy using LASSO: Loop based Automated Serial Sectioning Operation." *PLoS One*, 13.10, 2018.

Workshop Papers & Peer Reviewed Abstracts

Balwani A. "MechInterp for Recurrent Computation: Time-Resolved Circuit Discovery in RNNs." (Virtual Poster), *Mechanistic Interpretability Workshop, ICML*, 2026.

Efstathiou A., & **Balwani A.** "Recursive Reasoning as Attractor Landscape Search: Mechanistic Dynamics of the Tiny Recursive Model", *Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning (LIT), ICLR*, 2026.

Yusuf S., Nirvan S., & **Balwani A.** "RT-TopKSAE: Improving Top-k Sparse Autoencoders with the Rotation Trick.", *Workshop on Representational Alignment (Re-Align), ICLR*, 2026.

Kota M., Fan S., Poojary H., Teklehaimanot N., & **Balwani A.** "Novelty-Gated Experience Sharing for Multi-Agent Reinforcement Learning." (Poster), *Workshop on Multi-Agent Learning and its Opportunities in the Era of Generative AI (MALGAI), ICLR*, 2026.

Balwani A. "Time-Resolved Circuit Discovery in RNNs via Windowed Causal Interventions and Local Linearization." (Poster), *Symmetry and Geometry in Neural Representations (NeurReps), NeurIPS*, 2025. Also accepted as "Mechanistic Interpretability for Time-Resolved Neural Circuit Discovery in RNNs." (Poster), *COSYNE*, 2026.

Rahman A., Gurugubelli A., Ankit O., Zhu K., & **Balwani A.** "Probing the Origins of Reasoning Performance: Representational Quality for Mathematical Problem-Solving in RL vs SFT Finetuned Models." (Poster), *XAI4Science Workshop, AAAI*, 2026.

Arturi D., Zhang E., Ansah A., Zhu K., Panda A., & **Balwani A.** "Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior." (Honorable Mention – Top 4% and Oral – Top 7% of submissions), *UniReps Workshop, NeurIPS*, 2025. Also accepted at *Mechanistic Interpretability Workshop, NeurIPS*, 2025 (Spotlight Talk).

Durai A., Hu J., Buch K., Zhu K., Sharma V., & **Balwani A.** "LoRA-Guided PPO for Cost-Aware and Compute-Efficient Agent Orchestration." *Efficient Reasoning Workshop, NeurIPS*, 2025.

Balwani A., Wang A., Najafi F., & Choi H. "Constructing Biologically-Constrained RNNs via Dale's Backprop and Topologically-Informed Pruning." (Poster), *COSYNE*, 2025.

Zhou W., **Balwani A.**, Chung S., & Schneider D. "Motor-sensory Experience Reshapes Neural Manifolds in Auditory Cortex to Reflect Acoustic Expectations." *Advances and Perspectives in Auditory Neuroscience (APAN)*, 2023.

Balwani A., & Choi H. "On the Architectural Biases of the Canonical Cortical Microcircuit." (Talk, Top 3.2% of submissions), *COSYNE*, 2023.

Cho S., **Balwani A.**, & Choi H. "Leveraging Predictive Coding to Improve Artificial Neural Network Performance." (Poster), *Collaborative Research in Computational Neuroscience (CRCNS)*, 2022.

Balwani A., & Krzyston J. "Zeroth-order Topological Insights into Magnitude-based Neural Network Pruning." (Spotlight, Top 9.8% of submissions), *Topology, Algebra, and Geometry in Machine Learning, ICML*, 2022. Also presented as a poster at *Sparsity in Neural Networks*, 2022.

Balwani A., & Dyer E. "Modeling Brain Microarchitecture with Deep Representation Learning." (Poster), *ML Interpretability for Scientific Discovery, ICML*, 2020.

Balwani A., Miano J., Prasad J., & Dyer E. "Learning to Segment at Multiple Scales." (Poster), *BioImage Informatics*, 2019.

Milligan K., **Balwani A.**, Maguire A., Margulies S., & Dyer E. "Deep Learning for Characterization of Neuroinflammation in Traumatic Brain Injury." (Poster), *BioImage Informatics*, 2019.

Preprints

Sourav S., & **Balwani A.** "The Tell-Tale Trace: Detecting Reasoning Failures in LLMs Using Chain-of-Thought Dynamics." *arXiv*, 2026.

Amarnath C., **Balwani A.**, Ma K., & Chatterjee A. "TESDA: Transform Enabled Statistical Detection of Attacks in Deep Neural Networks." *arXiv*, 2021.

Balwani A., & Dyer E. "A Deep Feature Learning Approach for Mapping the Brain's Microarchitecture and Organization." *bioRxiv*, 2020.

Honours & Awards

Grants & Acquired Funding

- SPAR Project Funding, Fall 2026.
- BlueDot Impact Rapid Grant, August 2026.
- Open Philanthropy, Career Development and Transition Funding, 2025.

Academic Awards & Fellowships

- ECE Coulter MS Fellowship, Georgia Institute of Technology, 2016–2017.

Registration & Travel Awards

- Conference Registration Award, NeurIPS, 2025 (Sponsored by New Theory).
- COSYNE Presenters Travel Award, 2023.
- ICML Diversity and Inclusion Fellowship, 2020.

Competitions & Hackathons

- Best Poster – Frontiers in Science Conference and Symposium (Intelligence), Georgia Tech, 2025.
- Winner (Technical Track) – Hacklytics, Data Science at Georgia Tech, 2019.
- Winner (Best Project) – AI/ML for Social Good Hackathon at Georgia Tech, 2018.
- Gold Award – IEEE UBTech-Education Robotics Design Challenge, 2017.

Teaching & Mentoring

Supervised Program on Alignment Research (SPAR)

- Mentoring ∼8 mentees spanning high schoolers to masters students, working on projects studying unintended generalization in LLMs, through the lens of parametric mechanistic interpretability.

Algoverse AI Research

- Mentored ∼8 groups (>30 mentees) spanning high schoolers to post-PhD professionals, producing 8 accepted papers at NeurIPS, ICLR, ICML, and AAAI in mechanistic interpretability, representation learning, AI alignment/safety, and efficient learning.

Teaching Assistant

- Linear Algebra, Georgia Tech (Spring 2024)
- AI Safety Fundamentals, Georgia Tech (Facilitator, AI Safety Institute) (2023)
- Professional and Technical Communications for ECE, Georgia Tech (Summer 2021)
- Data Analytics for Engineers, Georgia Tech (Fall 2019, 2018)
- Hands-On Tech Day Camp, Georgia Tech (June 2019)
- Deep Learning for Microscopy Image Analysis, Marine Biological Laboratory (May 2019)
- Mathematical Foundations for Data Science, Georgia Tech (Spring 2018)
- Embedded Systems & IoT, Eduvance (Summer 2016)

Professional Service

Reviewing

- Journals: Nature Scientific Reports, PLoS One, Distill
- Conferences: NeurIPS, ICLR, COLM, AISTATS, MIDL, CoLLAs
- Workshops: Workshops at premier ML conferences – ICML, ICLR, NeurIPS, CVPR, etc. E.g., Workshop on Weight-Space Symmetries, Workshop on Geometrical and Topological Representation Learning, Topological Data Analysis and Beyond, Lifelong Learning Workshop, Workshop on Continual Learning in Computer Vision, Workshop on Continual Semi-Supervised Learning
- Other: Neuromatch Academy 2020, President's Undergraduate Research Awards (PURA), Georgia Tech

Professional & Student Organizations

- Senator (ECE), Graduate Student Association, Georgia Institute of Technology, 2017–2018

Workshops & Seminars

Attendee

- Define, Design, and Align, AI Safety @ UCLA (January 2023)
- AI Safety Workshop, Berkeley (December 2022)
- London Geometry and Machine Learning Summer School (July 2021)
- Banach Center – Oberwolfach Graduate Seminar: Mathematics of Deep Learning, Institute of Mathematics, Polish Academy of Sciences (November 2019)
- Foundation of Data Science Summer School, Georgia Institute of Technology (August 2019)
- Spinning Up in RL Workshop, OpenAI (February 2019)